

Predicting the causal relationship between polyunsaturated fatty acids and cerebral aneurysm risk from a Mendelian randomization study

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 1

March 5, 2024 (this version)

Posted to preprint server

November 16, 2023

Sent for peer review

November 15, 2023

Weijie Yu, Yunyun Mei, Chongfei Li, Zhenwei Lu, Xiaoyu Chen, Hao Yu, Xiaoyan Chen, Qionghui Huang, Zhangyu Li, Deyong Xiao, Liwei Zhou , Zhanxian Wang 

The School of Clinical Medicine, Fujian Medical University, Fuzhou 350122, China • Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China • Department of Neurosurgery, Fudan University Shanghai Cancer Center Xiamen Hospital, Xiamen 361026, China • School of Medicine, Xiamen University • Internal medicine, The First Affiliated Hospital of Xiamen University, Xiamen 361003, China • Department of Anesthesiology, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Introduction

No conclusive evidence for a link between polyunsaturated fatty acids (PUFA) and cerebral aneurysm has been found in observational research. The aim of our study was to determine the causal impact of PUFA on cerebral aneurysm.

Methods

Two sample Mendelian randomization (MR) was performed using genetic instruments derived from a recent genome wide association study (GWAS) of fatty acids from UK Biobank and outcome data obtained from the large-scale cerebral aneurysm GWASs in European ancestry which include IA, aneurysmal subarachnoid hemorrhage (aSAH) and unruptured intracranial aneurysm (uIA). Sensitivity analyses were implemented with MR-Egger intercept test, MR-PRESSO, leave-one-out analysis and so on. Bayesian colocalization (COLOC) methods was conducted to focus on the association between the fatty acid gene expression and cerebral aneurysm.

Results

Genetically predicted assessed omega-3 fatty acids decreased the risk for IA (OR = 0.80, 95% CI: 0.69 - 0.91, $P = 1.01 \times 10^{-3}$) and aSAH (OR = 0.71, 95% CI: 0.61 - 0.84, $P = 3.73 \times 10^{-5}$). Furthermore, the Docosahexaenoic acid decreased the risk for IA (OR = 0.75, 95% CI: 0.63 - 0.87, $P = 3.12 \times 10^{-4}$) and aSAH (OR = 0.67, 95% CI: 0.55 - 0.8, $P = 2.32 \times 10^{-5}$). The same results

were discovered from ratio of omega-3 fatty acids to total fatty acids. While the ratio of omega-6 fatty acids to omega-3 fatty acids increased the risk of IA (OR = 1.27, 95% CI: 1.12 – 1.44, $P = 1.53 \times 10^{-4}$) and aSAH (OR = 1.35, 95% CI: 1.17 – 1.56, $P = 5.78 \times 10^{-5}$). The result of the COLOC suggested that the above four kinds of fatty acids and IA, aSAH likely share causal variants in gene fatty acid desaturase 2, separately.

Conclusion

This study utilized integrative analysis of MR and colocalization to discover causal relationships between genetic variants, PUFA and cerebral aneurysm.

Funding

This study was funded by the Natural Science Foundation of China (82072777), the Natural Science Foundation of Xiamen (3502Z20227097), Fujian Provincial Health Commission, Provincial Health and Health Young and Middle-aged Backbone Talent Training Project (2022GGB010).

eLife assessment

Yu and colleagues used two-sample MR to test the effect of PUFA on cerebral aneurysms. They found that genetically predicted omega-3 and DHA decreased the risk for Intracranial Aneurysm and Subarachnoid Haemorrhage. This work is **useful**, although the evidence is **incomplete** as it lacks a robust set of analyses to provide credibility to the findings.

Introduction

Intracranial aneurysm (IA) is confined, pathological dilatations of the walls of intracranial arteries, which currently considered a chronic inflammatory disease that affects the intracranial arteries (Aoki et al., 2017 [\[1\]](#)) with a prevalence of approximately 3% in the general population (Vlak et al., 2011 [\[2\]](#)). About 85% of spontaneous subarachnoid hemorrhage (SAH) is due to ruptured IA (Macdonald et al., 2017 [\[3\]](#)). Unruptured IAs (uIAs) are frequently asymptomatic, but IA rupture can result in aneurysmal subarachnoid hemorrhage (aSAH), which has a poor prognosis (30% of death, 30% of independence and 30% of dependence), with patients often suffering from a disability or even death (Chen et al., 2020 [\[4\]](#); (Nieuwkamp et al., 2009 [\[5\]](#)). In light of the elevated rates of mortality and morbidity brought on by IA, it is crucial to explore the pathogenesis of intracranial aneurysm.

Polyunsaturated fatty acids (PUFA) are fatty acids that contain multiple double bonds in their molecular structure. According to the position and number of double bonds, polyunsaturated fatty acids can be further divided into two categories, omega-3 and omega-6. The long-chain omega-3 polyunsaturated fatty acid (omega-3) and docosahexaenoic acid (DHA) have been suggested to have cardioprotective (Burr et al., 1989 [\[6\]](#); (Wang et al., 2006 [\[7\]](#)), anti-inflammatory (Massaro et al., 2006 [\[8\]](#)), immunoregulatory (Yang et al., 2013 [\[9\]](#)), antioxidant (Carrepeiro et al., 2011 [\[10\]](#); (Bouzidi et al., 2010 [\[11\]](#)) and anti-tumor activities (Calviello et al., 2006 [\[12\]](#)). A recent study reported a significant decrease in red blood cell distribution width, pulse wave analysis and heart rate in patients with

abdominal aortic aneurysm who were supplemented with 1.8 g of long-chain omega-3 polyunsaturated fatty acids for 12 weeks, suggesting an improving in intravascular inflammation and vascular stiffness (Meital et al., 2019 [DOI](#); (Meital et al., 2020 [DOI](#)). Omega-3 and DHA down-regulate multiple aspects of the inflammatory process, such as oxidative stress and inflammation in macrophages (Meital et al., 2019 [DOI](#)). According to a recent retrospective study, high levels of free fatty acids on admission can provide some reference predictions for the poor prognosis of patients with aSAH at 3 months (Wang et al., 2021 [DOI](#)). Other studies demonstrated that arachidonic acid (AA) and AA-containing phospholipids were not detected in unruptured IA walls, whereas AA and AA-containing phospholipids were detected in significantly higher amounts in ruptured IA walls than in unruptured IA walls (Takeda et al., 2021 [DOI](#)).

Mendelian randomization (MR) is the use of genetic variation in non-experimental data to estimate the causal link between exposure and outcome, and it can reduce the impact of behavioral, social, psychological, and other factors (Davey Smith et al., 2014 [DOI](#)). And in recent years, many MR studies have emerged to provide clinical evidence (Hu et al., 2023 [DOI](#); (Yuan et al., 2019 [DOI](#)). To investigate the etiology of cerebral aneurysms, it is essential to further clarification of the causal relationship between PUFAs and cerebral aneurysms through Mendelian randomization.

Methods

Study design

In this study, we performed a two-sample Mendelian randomization analysis to examine the causal effects of PUFA and cerebral aneurysms using genome wide association study (GWAS) summary statistics. This instrumental-variable analysis mimics RCT with respect to the random allocation of single nucleotide polymorphisms (SNPs) in offspring (independent of confounding factors such as sex and age). In addition, this MR design has to fulfill three assumptions: (i) genetic instruments predict the exposure of interest ($P < 5 \times 10^{-8}$); (ii) genetic instruments are independent of potential confounders; (iii) genetic instruments affects the outcome only via the risk factors (Boef et al., 2015 [DOI](#)). The MR analysis used summary GWAS data publicly available from GWASs and no individual level data used. Due to the fact that each of the original GWASs had obtained ethical approval and participant consent, they were not required.

Data source and selection of genetic instruments

SNPs associated with 5 PUFAs were obtained from the open GWAS website, containing omega-3 fatty acids, omega-6 fatty acids, DHA, the ratio of omega-3 fatty acids to total fatty acids and ratio of omega-6 fatty acids to omega-3 fatty acids. According to the GWAS id, we named the ratio of omega-3 fatty acids to total fatty acids and ratio of omega-6 fatty acids to omega-3 fatty acids abbreviated as omega-3-pct and omega-6 by omega-3, respectively. They are derived from a large-scale UK Biobank GWAS of fatty acids with 114,999 individuals of European ancestry (Borges et al., 2022 [DOI](#)). Circulating omega-3 (i.e., DHA and total omega-3) and omega-6 (i.e., linoleic acid and total omega-6) fatty acid concentration were measured using a targeted high-throughput nuclear magnetic resonance metabolomics platform (Julkunen et al., 2021 [DOI](#)). The mean concentration of total omega-3 fatty acids, DHA and omega-6 fatty acids was 0.53 (SD 0.22) mmol/L, 0.23 (SD 0.08) mmol/L and 4.45 (SD 0.68) mmol/L. (Supplementary Table 1)

For the outcome, the genetic effect of the corresponding SNP on IA and aSAH were obtained from the most recent GWAS with a total of 65796 Europeans (6252 IA cases and 59544 controls) and 63740 Europeans (4196 aSAH cases and 59544 controls) after excluding UK Biobank (Bakker et al., 2020 [DOI](#)). And Summary statistics for uIA phenotypes were extracted from the from the FinnGen database (https://r8.finnngen.fi/pheno/I9_ANEURYSM [DOI](#)) with a total of 314556 Europeans (1788 uIA cases and 312768 controls). (Supplementary Table 1)

According three essential model assumptions of MR. First, SNPs were associated with the appropriate exposure at the genome-wide significance threshold $P < 5 \times 10^{-8}$. Second, to quantify the strength of the genetic instruments, SNPs with an F statistic less than 10 were excluded. Third, we clustered SNPs in linkage disequilibrium (LD, $R^2 \geq 0.001$ and within 10 mb). Fourth, to eliminate unclear, palindromic and associated with known confounding factors (body mass index (McDowell et al., 2018), blood pressure (Sun et al., 2022), type 2 diabetes (Tian et al., 2022), high-density lipoprotein (Huang et al., 2018)) SNPs. Fifth, If an instrumental SNP for the exposure was not available from the outcome data set, we replaced it with a suitable proxy SNP ($r^2 > 0.8$ in the European 1000 Genomes Project reference panel using LDlink <https://ldlink.nci.nih.gov/>) or removed it in the absence of such a proxy. SNPs located in the MHC region were removed. Finally, removing SNPs with potential pleiotropy and outlier through mendelian randomization pleiotropy residual sum and outlier (MR-PRESSO). Then, the heterogeneity between-SNP were tested using inverse variance weighting (IVW) and the MR-Egger method based on the SNPs that were retained after pleiotropy correction. (Figure 1) Details of the SNPs used as instrumental variables were displayed in Supplementary Tables 2 - 4.

Mendelian randomization analyses and Sensitivity analysis

Eleven MR analytical approaches were used in this study to investigate the causal impacts of PUFA on the cerebral aneurysm. As the main analysis, we used standard IVW estimates, is characterized by an analysis that does not take into account the presence of an intercept term and uses the inverse of the outcome variance (quadratic of the standard error) as a weight to provide a comprehensive estimate of the impact of the PUFA on the cerebral aneurysm. Furthermore, additional MR analyses, such as MR-Egger regression, weighted median, simple mode, weighted mode methods and so on were implemented as complements to the IVW, because these methods could provide more robust estimates over a wider range of scenarios. The MR-Egger method can provide causal estimates that are unaffected by breaches of standard IV assumptions and can detect whether standard IV assumptions are violated (Bowden et al., 2015). The weighted median method provides consistent estimates of effect, when at least half of the weighted variance provided by horizontal pleiotropy is valid (Bowden et al., 2016). We also performed reverse MR studies, where the cerebral aneurysms were treated as the exposure and PUFA was treated as the result, in order to thoroughly investigate the potential existence of reverse causation. To assess the robustness of the above results, a series of sensitivity analyses, including IVW, MR-Egger, MR-Egger intercept test, MR-PRESSO global test and “leave-one-out” method were conducted.

Bayesian colocalization (COLOC)

Bayesian colocalization (COLOC) analysis assesses whether single-nucleotide variants linked to gene expression and phenotype at the same locus are shared causal variants, and thus, gene expression and phenotype are “colocalized.” This approach assumes that: (1) in each test region, there exists at most one causal SNP for either trait; (2) the probability that a SNP is causal is independent of the probability that any other SNP in the genome is causal; (3) all causal SNPs are genotyped or imputed and included in analysis. According to these assumptions, there are five mutually exclusive hypotheses for each test region: (1) there is no causal SNP for either trait (H_0); (2) there is one causal SNP for trait 1 only (H_1); (3) there is one causal SNP for trait 2 only (H_2); (4) there are two distinct causal SNPs, one for each trait (H_3); and (5) there is a causal SNP common to both traits (H_4). A large PP for H_4 (PP. H_4 above 0.75) strongly supports shared causal variants affecting both gene expression and phenotype (Kurki et al., 2023).

Statistical Analysis

For the MR analysis, IVs with F values over 10 were considered strong instruments that could alleviate bias from weak instruments (Pierce et al., 2011). The formula used to calculate F values is as follows: $F = [(N-K-1) R^2] / [k(1-R^2)]$, where R^2 represents the proportion of variance explained by the genetic variants, N represents the sample size, and k represents the number of included

SNPs. $R^2 = \sum_1^k 2\beta^2(1 - EAF)EAF$, where EAF is the effect allele frequency and β is the estimated effect on LTL (Park et al., 2010). Given the limited number of cases in the GWAS, we calculated the statistical power for MR analysis using the mRnd website (<https://shiny.cnsgenomics.com/mRnd/>). To account for multiple testing in our primary analyses, a Bonferroni-corrected threshold of $P < 0.0033$ ($\alpha = 0.05/15$ [5 exposures and 3 outcomes]) was applied. All statistical analyses performed in this investigation were carried out using the “TwoSampleMR”, “phenoscanner”, “RadialMR”, “MendelianRandomization” and “MRPRESSO” packages in R software (version 4.2.2).

Results

According to the above methodology, we filtered out the corresponding SNPs. (Supplementary Table 2 – 4)

In the outcome of IA, IVW analysis indicated that omega-3 fatty acid (OR = 0.80, 95% CI: 0.69 - 0.91, $P = 1.01 \times 10^{-3}$), DHA (OR = 0.75, 95% CI: 0.63 - 0.87, $P = 3.12 \times 10^{-4}$) and omega-3-pct (OR = 0.8, 95% CI: 0.71 - 0.90, $P = 2.34 \times 10^{-4}$) decreased the risk for IA. And omega-6 by omega-3 increased the risk for IA (OR = 1.27, 95% CI: 1.12 - 1.44, $P = 1.53 \times 10^{-4}$). The results and directionality of other MR methods revealed consistency. (Supplementary Figure 1) However, there was no any causal relationship between Omega-6 fatty acid and IA. (Figure 2)

In the outcome of aSAH, omega-3 fatty acid (OR = 0.71, 95% CI: 0.61 - 0.84, $P = 3.73 \times 10^{-5}$), DHA (OR = 0.67, 95% CI: 0.55 - 0.80, $P = 2.32 \times 10^{-5}$) and omega-3-pct (OR = 0.76, 95% CI: 0.66 - 0.87, $P = 1.16 \times 10^{-4}$) decreased the risk for aSAH. And omega-6 by omega-3 increased the risk for aSAH (OR = 1.35, 95% CI: 1.17 - 1.56, $P = 5.78 \times 10^{-5}$). The results and directionality of other MR methods revealed consistency. (Supplementary Figure 2) However, there was no any causal relationship between omega-6 fatty acid and aSAH. (Figure 3)

In the outcome of uIA, there was no any causal relationship between all of the above five PUFAs and uIA. (Figure 4) In the result of reverse MR, MR analysis of all groups failed sterling filiting test.

To assess the robustness of the above results, a series of sensitivity analyses, including MR-Egger, IVW, MR-Egger intercept test, “leave-one-out” method and MR-PRESSO global test, were conducted (Supplementary Table 5 and Supplementary Figure 3-5). All P-values of the MR-Egger intercept tests, Q test and MR-PRESSO were > 0.05 , indicating that no horizontal pleiotropy and heterogeneity existed. All steiger test indicated correct directionality.

Although there is consistent evidence that PUFAs have a beneficial causal effect on cerebral aneurysm, it is uncertain if PUFAs are mediators in the hereditary causative route of cerebral aneurysm. For achieve that, we tested for shared causative SNPs between PUFAs and cerebral aneurysm using COLOC. In COLOC analysis, we further calculated the posterior probability of each SNP being causal to both of the traits (PPH4). We identified shared causal SNP of omega-3 fatty acid and IA, aSAH in gene fatty acid desaturase 2 (*FADS2*), respectively. Similarly, in DHA, omega-3-pct, omega-6 by omega-3, these three fatty acids and IA, aSAH also obtain the same result. (Table 1.)

Discussion

PUFA supplementation is controversial for the prevention of cerebral aneurysm disease, while cerebral aneurysm is a heterogeneous disease with various pathogenic mechanisms. Here, we implemented multiple rigorous MR methods to appraise the possible causal association of PUFA

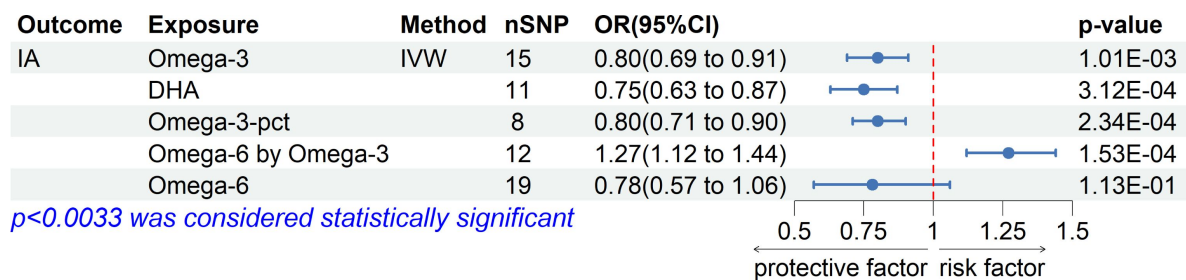


Figure 2.

Mendelian randomization analysis of the association between PUFA and IA.

PUFA: polyunsaturated fatty acids; IA: intracranial aneurysm; nSNP: number of single-nucleotide polymorphisms; OR: odds ratio; CI: confidence interval; MR, Mendelian randomization; IVW, inverse variance weighting; DHA: docosahexaenoic acid.

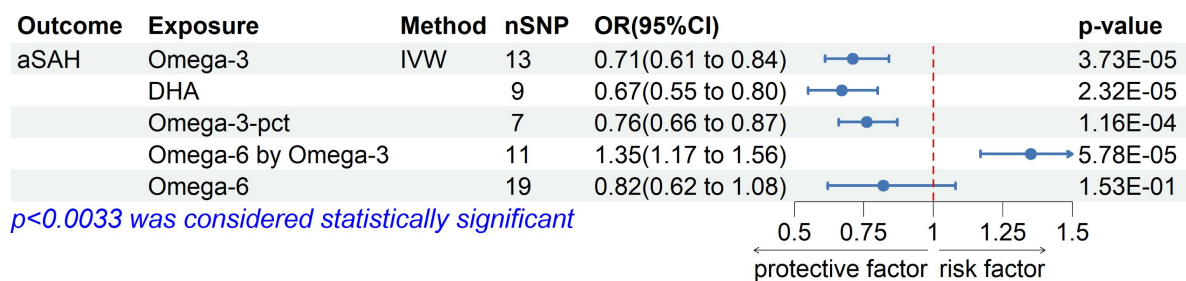


Figure 3.

Mendelian randomization analysis of the association between PUFA and aSAH.

PUFA: polyunsaturated fatty acids; aSAH: aneurysmal subarachnoid hemorrhage; nSNP: number of single-nucleotide polymorphisms; OR: odds ratio; CI: confidence interval; MR, Mendelian randomization; IVW, inverse variance weighting; DHA: docosahexaenoic acid.

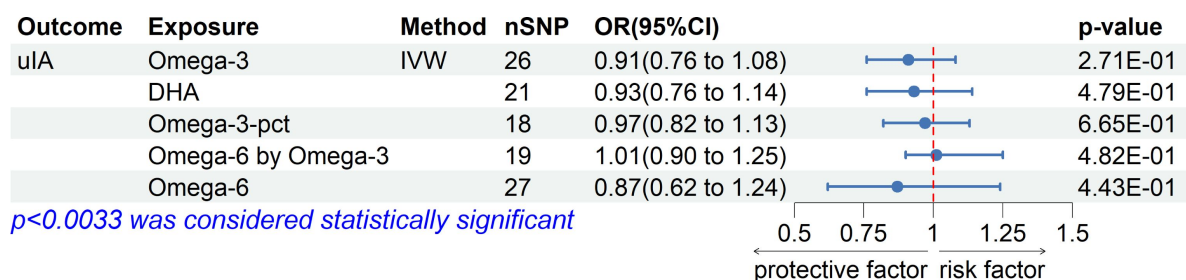


Figure 4.

Mendelian randomization analysis of the association between PUFA and uIA.

PUFA: polyunsaturated fatty acids; uIA: unruptured intracranial aneurysm; nSNP: number of single-nucleotide polymorphisms; OR: odds ratio; CI: confidence interval; MR, Mendelian randomization; IVW, inverse variance weighting; DHA: docosahexaenoic acid.

with cerebral aneurysm and its different states. We demonstrated that the omega-3 fatty acids, DHA, omega-3-pct, and omega-6 by omega-3 causally decreased the risk for IA and aSAH. And omega-6 fatty acids causally increased the risk of IA and aSAH. In addition to IVW, consistency was seen in the outcomes and directionality of different MR methods. We identified shared causal SNPs of omega-3 fatty acid, DHA, omega-3-pct, omega-6 by omega-3 and IA in genes *FADS2*, respectively. The same results were acquired from these four PUFAs and aSAH. The results of reverse MR and COLOC jointly increase the validity of our findings.

This could be connected to PUFAs can decrease the aneurysm's inflammatory process. The pathophysiology of IA involves inflammatory processes (Ge et al., 2022). which may include degeneration of the endothelium and vasa vasorum, as well as fragmentation of the internal elastic lamina (Nakatomi et al., 2000; Meng et al., 2014). As an underlying mechanism, the prostaglandin E2 in macrophages was considered by Aoki et al. as a factor regulating such chronic inflammation involved in the progression and rupture of IAs (Aoki et al., 2017). However, multiple elements of the inflammatory process, including oxidative stress and inflammation in macrophages, can be downregulated by omega-3 and DHA (Meital et al., 2019). In the Abdominal aortic aneurysm (AAA), supplementation with the omega-3 PUFA, DHA, suppressed production of 8-isoprostane, the gold standard biomarker of oxidative stress and a specific marker of lipid peroxidation (Kadiiska et al., 2005). The results concord with data from both experimental animal studies (Saraswathi et al., 2007; Zúñiga et al., 2010) and human clinical trials (Mori et al., 2003; Barden et al., 2015) that suggest omega-3 PUFAs decrease 8-isoprostane levels and ease oxidative stress. The findings from a randomized controlled trial suggest that omega-3 PUFAs improve erythrocyte fatty acid profile and ameliorate factors associated with inflammation in AAA patients (Meital et al., 2019). For aSAH, a prospective study demonstrated that administration of omega-3 fatty acids after aneurysmal SAH may reduce the frequency of cerebral vasospasm through the effects of anti-inflammatory and reduce both platelet aggregation and plasma triglycerides, and have neuroprotective effects and may improve clinical outcomes (Nakagawa et al., 2017). At the same time, it may have a preventive effect on the complication of intracranial hemorrhage after aneurysm surgery (Saito et al., 2017). A multi-center study with a sizable sample size is still required to clarify the trend of the effect of omega-3 fatty acids on aSAH, even though the favorable outcomes of these two studies have a certain reference value. Furthermore, Ririko et al. findings that AA and AA-containing phospholipids were not detected in the unruptured IA while they were detected in a significantly larger amount in the ruptured IA and suggest that the stability of the turnover of AA and linoleic acid in human unruptured IA walls is sustained (Takeda et al., 2021). Several earlier studies employing rat model found that the thickened aneurysmal wall and various brain areas after subarachnoid hemorrhage had higher amounts of free fatty acids, including AA (Gewirtz et al., 1999; Ikedo et al., 2018). This is in line with results of our MR analyses between omega-6 fatty acid and aSAH.

Besides, in the development of cardiovascular disease, genetically higher plasma α -linolenic and linoleic are inversely associated with large-artery stroke and venous thromboembolism, whereas AA level have a favorable correlation with these cardiovascular diseases (Yuan et al., 2019). This is may be associated with the $\Delta 5$ -desaturase and $\Delta 6$ -desaturase, which are the primary rate-limiting enzyme for the synthesis of inborn long-chain PUFAs and is encoded by the fatty acid desaturase 1 (*FADS1*) gene and fatty acid desaturase 2 (*FADS2*) gene (Simopoulos, 2010). Even the aortic valve's fatty acid composition might be impacted by *FADS1* (Plunde et al., 2020). In our COLOC analysis, we identified shared causal SNP of omega-3 fatty acid and IA, aSAH in gene fatty acid desaturase 2 (*FADS2*), respectively. The same result was also found in DHA, omega-3-pct, omega-6 by omega-3. It is obvious that omega-3 fatty acids play a vital role in vascular-related diseases. Conversely, it's incorrect to assume that all forms of omega-6 fatty acids would worsen blood vessel damage. More thorough large-sample study is still required for additional validation.

However, several limitations should be taken into account in our study. First, given that the participants in this study were from European population, the results of this study cannot be smoothly extrapolated to other ethnic groups with diverse lifestyles and cultural backgrounds. Hence, further multi-ancestry studies are needed to evaluate whether our findings can be generalized to individuals of other ancestries. Second, since MR analyses extrapolated causal hypotheses by exploiting the random allocation of genetic variants, it was difficult to completely distinguish between mediation and pleiotropy using MR approaches. The generous variants in our genome probably affect one or more phenotypes, even though we made an effort to remove the impacts of pleiotropic and heterogeneous factors and confounders, some effects may still be there. Third, not all SNPs were examined, as some were removed because of LD (and no proxy SNPs were found), which may have impacted the results. Fourth PUFAs were measured from nonfasting peripheral plasma specimens, not aneurysm cavity blood and aneurysm wall tissue. Fifth, there is currently a lack of GWAS datasets of cerebral aneurysms from different cerebrovascular sites, and exploring the relationship between genetically predicted PUFAs and the occurrence of specific types of cerebral aneurysms remains a challenge in current MR analyses.

Conclusion

In view of the extremely poor prognosis of cerebral aneurysms and the high mortality rate of secondary rupture, our research focuses on the causal relationship between PUFAs and cerebral aneurysms to improve our understanding of the pathogenic mechanism of various PUFAs in cerebral aneurysms. And provide new insights into the diagnosis and treatment of brain aneurysms in the future.

Data Availability

The GWAS databases we use are all from publicly published data, each of the original GWASs had obtained ethical approval and participant consent

Acknowledgements

The authors want to acknowledge all of the participants and investigators for contributing and sharing summary-level data on GWAS.

Consent for Publication

All participants provided informed consent before enrollment in our study.

Disclosures

The authors declare that they have no competing interests.

Author contributions

Study design: Weijie Yu, Yunyun Mei, Liwei Zhou and Zhanxiang Wang. Acquisition of datasets: Weijie Yu, Yunyun Mei, Chongfei Li, Xiaoyan Chen, Zhenwei Lu, Qionghui Huang and Zhangyu Li. Data analysis: Weijie Yu, Yunyun Mei and Chongfei Li. Manuscript drafting: Weijie Yu and Yunyun Mei. Manuscript revision: Liwei Zhou and Zhanxiang Wang. Final approval of manuscript: all authors.

Ethical Approval

Due to the fact that each of the original GWASs had obtained ethical approval and participant consent, they were not required.

Data availability

The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

References

- Aoki T, Frösen J, Fukuda M, Bando K, Shioi G, Tsuji K, Ollikainen E, Nozaki K, Laakkonen J, Narumiya S. (2017) **Prostaglandin E2-EP2-NF-κB signaling in macrophages as a potential therapeutic target for intracranial aneurysms** *Science signaling* **10**:465 <https://doi.org/10.1126/scisignal.aah6037>
- Vlak MH, Algra A, Brandenburg R, Rinkel GJ (2011) **Prevalence of unruptured intracranial aneurysms, with emphasis on sex, age, comorbidity, country, and time period: a systematic review and meta-analysis** *The Lancet Neurology* **10**:626–636 [https://doi.org/10.1016/s1474-4422\(11\)70109-0](https://doi.org/10.1016/s1474-4422(11)70109-0)
- Macdonald RL, Schweizer TA (2017) **Spontaneous subarachnoid haemorrhage** *The Lancet*
- Chen Y *et al.* (2020) **Mortality and recurrent vascular events after first incident stroke: a 9-year community-based study of 0.5 million Chinese adults** *The Lancet Global health* **8**:e580–e590 [https://doi.org/10.1016/s2214-109x\(20\)30069-3](https://doi.org/10.1016/s2214-109x(20)30069-3)
- Nieuwkamp DJ, Setz LE, Algra A, Linn FH, de Rooij NK, Rinkel GJ (2009) **Changes in case fatality of aneurysmal subarachnoid haemorrhage over time, according to age, sex, and region: a meta-analysis** *The Lancet Neurology* **8**:635–642 [https://doi.org/10.1016/s1474-4422\(09\)70126-7](https://doi.org/10.1016/s1474-4422(09)70126-7)
- Burr ML, Fehily AM, Gilbert JF, Rogers S, Holliday RM, Sweetnam PM, Elwood PC, Deadman NM (1989) **Effects of changes in fat, fish, and fibre intakes on death and myocardial reinfarction: diet and reinfarction trial (DART).** *Lancet (London England)* **286**:757–761 [https://doi.org/10.1016/s0140-6736\(89\)90828-3](https://doi.org/10.1016/s0140-6736(89)90828-3)
- Wang C, Harris WS, Chung M, Lichtenstein AH, Balk EM, Kupelnick B, Jordan HS, Lau J (2006) **n-3 Fatty acids from fish or fish-oil supplements, but not alpha-linolenic acid, benefit cardiovascular disease outcomes in primary- and secondary-prevention studies: a systematic review** *The American journal of clinical nutrition* **84**:5–17 <https://doi.org/10.1093/ajcn/84.1.5>
- Massaro M, Habib A, Lubrano L, Del Turco S, Lazzerini G, Bourcier T, Weksler BB, De Caterina R (2006) **The omega-3 fatty acid docosahexaenoate attenuates endothelial cyclooxygenase-2 induction through both NADP(H) oxidase and PKC epsilon inhibition** *Proceedings of the National Academy of Sciences of the United States of America* **103**:15184–15189 <https://doi.org/10.1073/pnas.0510086103>
- Yang YC, Lii CK, Wei YL, Li CC, Lu CY, Liu KL, Chen HW (2013) **Docosahexaenoic acid inhibition of inflammation is partially via cross-talk between Nrf2/heme oxygenase 1 and IKK/NF-κB pathways** *The Journal of nutritional biochemistry* **24**:204–212 <https://doi.org/10.1016/j.jnutbio.2012.05.003>
- Carrepeiro MM, Rogero MM, Bertolami MC, Botelho PB, Castro N, Castro IA (2011) **Effect of n-3 fatty acids and statins on oxidative stress in statin-treated hypercholesterolemic and normocholesterolemic women** *Atherosclerosis* **217**:171–178 <https://doi.org/10.1016/j.atherosclerosis.2010.12.013>

Bouzidi N, Mekki K, Boukaddoum A, Dida N, Kaddous A, Bouchenak M (2010) **Effects of omega-3 polyunsaturated fatty-acid supplementation on redox status in chronic renal failure patients with dyslipidemia** *Journal of renal nutrition : the official journal of the Council on Renal Nutrition of the National Kidney Foundation* **205**:321–328 <https://doi.org/10.1053/j.jrn.2010.01.002>

Calviello G, Serini S, Palozza P (2006) . **n-3 polyunsaturated fatty acids as signal transduction modulators and therapeutical agents in cancer** *Current Signal Transduction Therapy* **13**:255–271

Meital LT *et al.* (2019) . **n-3 PUFAs improve erythrocyte fatty acid profile in patients with small AAA: a randomized controlled trial** *Journal of lipid research* **606**:1154–1163 <https://doi.org/10.1194/jlr.P093013>

Meital LT, Schulze K, Magee R, O'Donnell J, Jha P, Meital CY, Donkin R, Bailey TG, Askew CD, Russell FD (2020) **Long Chain Omega-3 Polyunsaturated Fatty Acids Improve Vascular Stiffness in Abdominal Aortic Aneurysm: A Randomized Controlled Trial** *Nutrients* **131** <https://doi.org/10.3390/nu13010138>

Meital LT, Windsor MT, Perissiou M, Schulze K, Magee R, Kuballa A, Golledge J, Bailey TG, Askew CD, Russell FD (2019) **Omega-3 fatty acids decrease oxidative stress and inflammation in macrophages from patients with small abdominal aortic aneurysm** *Scientific reports* **91** <https://doi.org/10.1038/s41598-019-49362-z>

Wang Y *et al.* (2021) **Development and Clinical Translation of a Perioperative Nomogram Incorporating Free Fatty Acids to Predict Poor Outcome of Aneurysmal Subarachnoid Hemorrhage Following Endovascular Treatment** *Frontiers in neurology* **12** <https://doi.org/10.3389/fneur.2021.629997>

Takeda R, Islam A, Sato T, Kurita H, Kahyo T, Urano T, Setou M (2021) **The stability of the metabolic turnover of arachidonic acid in human unruptured intracranial aneurysmal walls is sustained** *Clinical neurology and neurosurgery* **208** <https://doi.org/10.1016/j.clineuro.2021.106881>

Smith G Davey, Hemani G (2014) **Mendelian randomization: genetic anchors for causal inference in epidemiological studies** *Human molecular genetics* **23** <https://doi.org/10.1093/hmg/ddu328>

Hu C *et al.* (2023) **Evaluating the distinct pleiotropic effects of omega-3 fatty acids on type 2 diabetes mellitus: a mendelian randomization study** *Journal of translational medicine* **211** <https://doi.org/10.1186/s12967-023-04202-7>

Yuan S, Bäck M, Bruzelius M, Mason AM, Burgess S, Larsson S (2019) **Plasma Phospholipid Fatty Acids, FADS1 and Risk of 15 Cardiovascular Diseases: A Mendelian Randomisation Study** *Nutrients* <https://doi.org/10.3390/nu11123001>

Boef AG, Dekkers OM, le Cessie S (2015) **Boef AG, Dekkers OM, le Cessie S. 2015. Mendelian randomization studies: a review of the approaches used and the quality of reporting. International journal of epidemiology. 442:496-511. DOI: 10.1093/ije/dyv071** *Mendelian randomization studies: a review of the approaches used and the quality of reporting. International journal of epidemiology* **442** <https://doi.org/10.1093/ije/dyv071>

Borges MC, Haycock PC, Zheng J, Hemani G, Holmes MV, Davey Smith G, Hingorani AD, Lawlor DA (2022) **Role of circulating polyunsaturated fatty acids on cardiovascular diseases risk: analysis using Mendelian randomization and fatty acid genetic association data from over 114,000 UK Biobank participants** *BMC medicine* **201** <https://doi.org/10.1186/s12916-022-02399-w>

Julkunen H, Cichońska A, Slagboom PE, Würtz P (2021) **Metabolic biomarker profiling for identification of susceptibility to severe pneumonia and COVID-19 in the general population** *eLife* **10** <https://doi.org/10.7554/eLife.63033>

Bakker MK *et al.* (2020) **Bakker MK, van der Spek RAA, van Rheenen W, Morel S, Bourcier R, Hostettler IC, Alg VS, van Eijk KR, Koido M, Akiyama M, Terao C, Matsuda K, Walters RG, Lin K, Li L, Millwood IY, Chen Z, Rouleau GA, Zhou S, Rannikmäe K, Sudlow CLM, Houlden H, van den Berg LH, Dina C, Naggara O, Gentric JC, Shotar E, Eugène F, Desal H, Winsvold BS, Børte S, Johnsen MB, Brumpton BM, Sandvei MS, Willer CJ, Hveem K, Zwart JA, Verschuren WMM, Friedrich CM, Hirsch S, Schilling S, Dauvillier J, Martin O, Jones GT, Bown MJ, Ko NU, Kim H, Coleman JRI, Breen G, Zaroff JG, Klijn CJM, Malik R, Dichgans M, Sargurupremraj M, Tatlisumak T, Amouyel P, Debette S, Rinkel GJE, Worrall BB, Pera J, Slowik A, Gaál-Paavola EI, Niemelä M, Jääskeläinen JE, von Und Zu Fraunberg M, Lindgren A, Broderick JP, Werring DJ, Woo D, Redon R, Bijslenga P, Kamatani Y, Veldink JH, Ruigrok YM.** 2020. **Genome-wide association study of intracranial aneurysms identifies 17 risk loci and genetic overlap with clinical risk factors.** *Nature genetics*. **5212:1303-1313**. DOI: **10.1038/s41588-020-00725-7** *Genome-wide association study of intracranial aneurysms identifies 17 risk loci and genetic overlap with clinical risk factors. Nature genetics* **5212** <https://doi.org/10.1038/s41588-020-00725-7>

McDowell MM, Zhao Y, Kellner CP, Barton SM, Sussman E, Claassen J, Ducruet AF, Connolly ES (2018) **Demographic and clinical predictors of multiple intracranial aneurysms in patients with subarachnoid hemorrhage** *Journal of neurosurgery* **1284:961–968** <https://doi.org/10.3171/2017.1.Jns162785>

Sun X, Liu B, Chen Y, Lv L, Ye D, Mao Y (2022) **Modifiable risk factors for intracranial aneurysms: Evidence from genetic studies** *International journal of stroke : official journal of the International Stroke Society* **1710:1107–1113** <https://doi.org/10.1177/17474930211065640>

Tian D, Zhang L, Zhuang Z, Huang T, Fan D (2022) **A two-sample Mendelian randomization analysis of modifiable risk factors and intracranial aneurysms** *Scientific reports* **121** <https://doi.org/10.1038/s41598-022-11720-9>

Huang Q, Shang-Guan HC, Wu SY, Yao PS, Sun Y, Zeng YL, Zheng SF, Chen GR, Lin YX, Kang DZ (2018) **High-Density Lipoprotein Is Associated with Progression of Intracranial Aneurysms** *World neurosurgery* **120:e234–e240** <https://doi.org/10.1016/j.wneu.2018.08.037>

Bowden J, Davey Smith G, Burgess S (2015) **Mendelian randomization with invalid instruments: effect estimation and bias detection through Egger regression** *International journal of epidemiology* **442:512–525** <https://doi.org/10.1093/ije/dyv080>

Bowden J, Davey Smith G, Haycock PC, Burgess S (2016) **Consistent Estimation in Mendelian Randomization with Some Invalid Instruments Using a Weighted Median Estimator** *Genetic epidemiology* **404:304–314** <https://doi.org/10.1002/gepi.21965>

Kurki MI *et al.* (2023) **FinnGen provides genetic insights from a well-phenotyped isolated population** *Nature* **6137944:508–518** <https://doi.org/10.1038/s41586-022-05473-8>

Pierce BL, Ahsan H, Vanderweele TJ (2011) **Power and instrument strength requirements for Mendelian randomization studies using multiple genetic variants** *International journal of epidemiology* **403**:740–752 <https://doi.org/10.1093/ije/dyq151>

Park JH, Wacholder S, Gail MH, Peters U, Jacobs KB, Chanock SJ, Chatterjee N (2010) **Estimation of effect size distribution from genome-wide association studies and implications for future discoveries** *Nature genetics* **427**:570–575 <https://doi.org/10.1038/ng.610>

Ge P *et al.* (2022) **High-Dimensional Immune Profiling by Mass Cytometry Revealed the Circulating Immune Cell Landscape in Patients With Intracranial Aneurysm** *Frontiers in immunology* **13** <https://doi.org/10.3389/fimmu.2022.922000>

Nakatomi H, Segawa H, Kurata A, Shiokawa Y, Nagata K, Kamiyama H, Ueki K, Kirino T (2000) **Clinicopathological study of intracranial fusiform and dolichoectatic aneurysms : insight on the mechanism of growth** *Stroke* **314**:896–900 <https://doi.org/10.1161/01.str.31.4.896>

Meng H, Tutino VM, Xiang J, Siddiqui A (2014) **High WSS or low WSS? Complex interactions of hemodynamics with intracranial aneurysm initiation, growth, and rupture: toward a unifying hypothesis.** *AJNR American journal of neuroradiology* **357**:1254–1262 <https://doi.org/10.3174/ajnr.A3558>

Kadiiska MB *et al.* (2005) **Biomarkers of oxidative stress study II: are oxidation products of lipids, proteins, and DNA markers of CCl4 poisoning?** *Free radical biology & medicine* **386** <https://doi.org/10.1016/j.freeradbiomed.2004.09.017>

Saraswathi V, Gao L, Morrow JD, Chait A, Niswender KD, Hasty AH (2007) **Fish oil increases cholesterol storage in white adipose tissue with concomitant decreases in inflammation, hepatic steatosis, and atherosclerosis in mice** *The Journal of nutrition* **1377**:1776–1782 <https://doi.org/10.1093/jn/137.7.1776>

Zúñiga J, Venegas F, Villarreal M, Núñez D, Chandía M, Valenzuela R, Tapia G, Varela P, Videla LA, Fernández V (2010) **Protection against in vivo liver ischemia-reperfusion injury by n-3 long-chain polyunsaturated fatty acids in the rat** *Free radical research* **448**:854–863 <https://doi.org/10.3109/10715762.2010.485995>

Mori TA, Woodman RJ, Burke V, Puddey IB, Croft KD, Beilin LJ (2003) **Effect of eicosapentaenoic acid and docosahexaenoic acid on oxidative stress and inflammatory markers in treated-hypertensive type 2 diabetic subjects** *Free radical biology & medicine* **357**:772–781 [https://doi.org/10.1016/s0891-5849\(03\)00407-6](https://doi.org/10.1016/s0891-5849(03)00407-6)

Barden AE, Burke V, Mas E, Beilin LJ, Puddey IB, Watts GF, Irish AB, Mori TA (2015) **n-3 fatty acids reduce plasma 20-hydroxyeicosatetraenoic acid and blood pressure in patients with chronic kidney disease** *Journal of hypertension* **339**:1947–1953 <https://doi.org/10.1097/hjh.0000000000000621>

Nakagawa I *et al.* (2017) **ω-3 Fatty Acids Ethyl Esters Suppress Cerebral Vasospasm and Improve Clinical Outcome Following Aneurysmal Subarachnoid Hemorrhage** *World neurosurgery* **99**:457–464 <https://doi.org/10.1016/j.wneu.2016.12.018>

Saito G, Zapata R, Rivera R, Zambrano H, Rojas D, Acevedo H, Ravera F, Mosquera J, Vásquez JE, Mura J (2017) **Long-chain omega-3 fatty acids in aneurysmal subarachnoid hemorrhage: A randomized pilot trial of pharmaconutrition** *Surgical neurology international* **8** https://doi.org/10.4103/sni.sni_266_17

Gewirtz RJ, Dhillon HS, Goes SE, DeAtley SM, Scheff SW (1999) **Lactate and free fatty acids after subarachnoid hemorrhage** *Brain research* **8401** [https://doi.org/10.1016/s0006-8993\(99\)01752-7](https://doi.org/10.1016/s0006-8993(99)01752-7)

Ikedo T, Minami M, Kataoka H, Hayashi K, Nagata M, Fujikawa R, Yamazaki F, Setou M, Yokode M, Miyamoto S (2018) **Imaging mass spectroscopy delineates the thinned and thickened walls of intracranial aneurysms** *Biochemical and biophysical research communications* **4951**:332–338 <https://doi.org/10.1016/j.bbrc.2017.10.133>

Simopoulos AP (2010) **Genetic variants in the metabolism of omega-6 and omega-3 fatty acids: their role in the determination of nutritional requirements and chronic disease risk** *Experimental biology and medicine (Maywood, NJ)* **2357**:785–795 <https://doi.org/10.1258/ebm.2010.009298>

Plunde O, Larsson SC, Artiach G, Thanassoulis G, Carracedo M, Franco-Cereceda A, Eriksson P, Bäck M (2020) **FADS1 (Fatty Acid Desaturase 1) Genotype Associates With Aortic Valve FADS mRNA Expression** *Fatty Acid Content and Calcification. Circulation Genomic and precision medicine* **133** <https://doi.org/10.1161/circgen.119.002710>

Article and author information

Weijie Yu

The School of Clinical Medicine, Fujian Medical University, Fuzhou 350122, China, Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

Yunyun Mei

Department of Neurosurgery, Fudan University Shanghai Cancer Center Xiamen Hospital, Xiamen 361026, China

Chongfei Li

The School of Clinical Medicine, Fujian Medical University, Fuzhou 350122, China, Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

Zhenwei Lu

The School of Clinical Medicine, Fujian Medical University, Fuzhou 350122, China, Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

Xiaoyu Chen

School of Medicine, Xiamen University

Hao Yu

School of Medicine, Xiamen University

Xiaoyan Chen

Internal medicine, The First Affiliated Hospital of Xiamen University, Xiamen 361003, China

Qionghui Huang

The School of Clinical Medicine, Fujian Medical University, Fuzhou 350122, China, Department of Anesthesiology, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

Zhangyu Li

Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

Deyong Xiao

Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

Liwei Zhou

Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

For correspondence: 1769958372@qq.com

Zhanxian Wang

Department of Neurosurgery, Department of Neurosurgery Quality Control Center, The First Affiliated Hospital of Xiamen University, Xiamen Key Laboratory of Brain Center, Xiamen 361003, China

For correspondence: zhanxiangW_sjwk@163.com

ORCID iD: [0009-0001-6101-7776](https://orcid.org/0009-0001-6101-7776)

Copyright

© 2024, Yu et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Edward Janus

University of Melbourne, Melbourne, Australia

Senior Editor

Eduardo Franco

McGill University, Montreal, Canada

Reviewer #1 (Public Review):

Summary:

The authors performed two-sample MR combined with sensitivity analyses and colocalization to test the effect of PUFA on cerebral aneurysms. They found that genetically predicted

omega-3 and DHA decreased the risk for intracranial aneurysm (IA) and subarachnoid haemorrhage (SAH) but not for unruptured IA (uIA).

Strengths:

PUFA on the risk of cerebral aneurysms is of clinical importance; the authors performed multiple sensitivity analyses to ensure MR fulfills its assumptions.

Weakness:

In my opinion, the major weakness is the selection of IVs, the same IVs should be used for each exposure, especially when the outcomes (IA, SAH, and uIA) are closely related. The removal of IVs was inconsistent, for example, why was LPA rs10455872 removed for SAH but not for uIA? (significantly more IVs were used for uIA). The authors should provide more details for the justification of the removal of IVs other than only indicating "confounder" in supplementary tables. The authors should also perform additional analyses including all IVs and IVs from other PUFA GWAS.

In addition, it seems that the SNPs in the FADS locus were driving the MR association, while FADS is a very pleiotropic locus associated with many lipid traits, removing FADS could attenuate the MR effect. The authors should perform a sensitivity analysis to remove this locus.

Instead of removing multiple "confounder" IVs which I think may bias the MR results due to very closely related lipid traits, the authors should perform multivariable MR to identify independent effects of PUFAs to IA, conditioning on other PUFAs and/or other lipids.

Colocalization was not well described, the authors should include the colocalization results for each locus in a supplementary table. They also mentioned "a large PP for H4 (PP.H4 above 0.75) strongly supports shared causal variants affecting both gene expression and phenotype". The authors should make sure that the colocalization was performed using the expression data of each gene or using the GWAS summary of each PUFA locus.

<https://doi.org/10.7554/eLife.94064.1.sa1>

Reviewer #2 (Public Review):

Summary:

In the manuscript, Yu et al reported a two-sample Mendelian randomization study to evaluate the causation between polyunsaturated fatty acids (PUFA) and cerebral aneurysm, based on summary statistics from published genome-wide association studies. The authors identified that omega-3 fatty acids and Docosahexaenoic acid decreased the risk for intracranial aneurysm (IA) and aneurysmal subarachnoid hemorrhage (aSAH). COLOC analysis suggested that the acids and IA, aSAH likely share causal variants in gene fatty acid desaturase 2.

Strengths:

The methodology is sound, with appropriate sensitivity analysis.

Weaknesses:

The results did not provide significant novel findings. The interpretation of the results is not sound.

<https://doi.org/10.7554/eLife.94064.1.sa0>